DEWAGEDARA D.M.E.S.
E/21/087

```
mysql>
mysql> SELECT * FROM students;
+------------+----------+------------+---------------------+------+
| student_id | name     | dob        | email               | gpa  |
+------------+----------+------------+---------------------+------+
|          6 | Uvindu   | 2002-03-20 | e21006@eng.pdn.ac.lk | NULL |
|         49 | Lahiru   | 2002-09-01 | e21049@eng.pdn.ac.lk | NULL |
|        138 | FIKRY    | 2002-01-15 | e21138@eng.pdn.ac.lk | 3.95 |
|        217 | Ishan    | 2002-06-21 | e21217@eng.pdn.ac.lk | 3.35 |
|        245 | Kaveesha | 2002-06-05 | e21245@eng.pdn.ac.lk | 3.98 |
|        302 | Sahandi  | 2002-05-27 | e21302@eng.pdn.ac.lk | 3.87 |
+------------+----------+------------+---------------------+------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM course_instructors;
+-----------+---------------+
| course_id | instructor_id |
+-----------+---------------+
|       224 |            10 |
|       225 |            11 |
|       226 |            12 |
|       214 |            13 |
+-----------+---------------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM instructors;
+---------------+----------------+-----------------------+
| instructor_id | name           | department            |
+---------------+----------------+-----------------------+
|            10 | Saadia Jameel  | Computer Engineering  |
|            11 | Shakthi Perera | Electrical Engineering |
|            12 | Ridma          | Computer Engineering  |
|            13 | Akila          | Computer Engineering  |
+---------------+----------------+-----------------------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM courses;
+-----------+----------------------+---------+
| course_id | course_name          | credits |
+-----------+----------------------+---------+
|       214 | Numerical Methods    |       3 |
|       215 | Calculas             |       2 |
|       224 | Computer Architecture |       4 |
|       225 | Software Construction |       3 |
|       226 | Database Systems     |       3 |
|       285 | Electronic           |       3 |
+-----------+----------------------+---------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM enrollments;
+------------+-----------+-------------+-------+
| student_id | course_id | enroll_date | grade |
+------------+-----------+-------------+-------+
|          6 |       214 | 2024-02-25  | B     |
|          6 |       215 | 2024-02-25  | B     |
|          6 |       224 | 2024-02-25  | B     |
|          6 |       225 | 2024-02-25  | B     |
|          6 |       226 | 2024-03-04  | A     |
|          6 |       285 | 2024-02-25  | B     |
|         49 |       214 | 2024-02-05  | C     |
|         49 |       215 | 2024-02-15  | NULL  |
|         49 |       224 | 2024-02-15  | NULL  |
|         49 |       225 | 2024-02-15  | C     |
|         49 |       226 | 2024-03-05  | NULL  |
|         49 |       285 | 2024-02-15  | NULL  |
|        138 |       214 | 2024-02-01  | A     |
|        138 |       215 | 2024-02-11  | A     |
|        138 |       224 | 2024-02-01  | A     |
|        138 |       225 | 2024-02-01  | A     |
|        138 |       226 | 2024-03-01  | A     |
|        138 |       285 | 2024-02-01  | A     |
|        217 |       214 | 2024-02-02  | B     |
|        217 |       215 | 2024-02-01  | B     |
|        217 |       224 | 2024-02-01  | B     |
|        217 |       225 | 2024-02-01  | B     |
|        217 |       226 | 2024-03-01  | B     |
|        217 |       285 | 2024-02-01  | B     |
|        245 |       214 | 2024-02-09  | A     |
|        245 |       215 | 2024-02-09  | A     |
|        245 |       224 | 2024-02-09  | A     |
|        245 |       225 | 2024-02-09  | A     |
|        245 |       226 | 2024-03-03  | A     |
|        245 |       285 | 2024-02-09  | A     |
|        302 |       214 | 2024-02-03  | A     |
|        302 |       215 | 2024-02-10  | A     |
|        302 |       224 | 2024-02-10  | A     |
|        302 |       225 | 2024-02-10  | A     |
|        302 |       226 | 2024-03-02  | B     |
|        302 |       285 | 2024-02-10  | A     |
+------------+-----------+-------------+-------+
```

Lab Tasks

1)

a. List all students details.

```
mysql>
mysql> SELECT * FROM students;
+------------+----------+------------+----------------------+------+
| student_id | name     | dob        | email                | gpa  |
+------------+----------+------------+----------------------+------+
|          6 | Uvindu   | 2002-03-20 | e21006@eng.pdn.ac.lk | NULL |
|         49 | Lahiru   | 2002-09-01 | e21049@eng.pdn.ac.lk | NULL |
|        138 | FIKRY    | 2002-01-15 | e21138@eng.pdn.ac.lk | 3.95 |
|        217 | Ishan    | 2002-06-21 | e21217@eng.pdn.ac.lk | 3.35 |
|        245 | Kaveesha | 2002-06-05 | e21245@eng.pdn.ac.lk | 3.98 |
|        302 | Sahandi  | 2002-05-27 | e21302@eng.pdn.ac.lk | 3.87 |
+------------+----------+------------+----------------------+------+
6 rows in set (0.00 sec)
```

Use the SELECT to show all information about the students stored in the table.

b. Display all courses that offer more than 3 credits.

```
mysql> SELECT * FROM courses where credits>3;
+-----------+----------------------+---------+
| course_id | course_name          | credits |
+-----------+----------------------+---------+
|       224 | Computer Architecture |       4 |
+-----------+----------------------+---------+
1 row in set (0.00 sec)
```

Use WHERE clause to filter and display only the courses that have more than 3 credits.

c. Show all enrollments with student name, course name, and grade.

```
mysql> SELECT s.name AS student_name, c.course_name, e.grade
    -> FROM enrollments e
    -> JOIN students s ON e.student_id = s.student_id
    -> JOIN courses c ON e.course_id = c.course_id;
```

Use JOIN to combine data from students, enrollments, and courses tables and display student name, course name, and grade together using SELECT.

student_name	course_name	grade
Uvindu	Numerical Methods	B
Uvindu	Calculas	B
Uvindu	Computer Architecture	B
Uvindu	Software Construction	B
Uvindu	Database Systems	A
Uvindu	Electronic	B
Lahiru	Numerical Methods	C
Lahiru	Calculas	NULL
Lahiru	Computer Architecture	NULL
Lahiru	Software Construction	C
Lahiru	Database Systems	NULL
Lahiru	Electronic	NULL
FIKRY	Numerical Methods	A
FIKRY	Calculas	A
FIKRY	Computer Architecture	A
FIKRY	Software Construction	A
FIKRY	Database Systems	A
FIKRY	Electronic	A
Ishan	Numerical Methods	B
Ishan	Calculas	B
Ishan	Computer Architecture	B
Ishan	Software Construction	B
Ishan	Database Systems	B
Ishan	Electronic	B
Kaveesha	Numerical Methods	A
Kaveesha	Calculas	A
Kaveesha	Computer Architecture	A
Kaveesha	Software Construction	A
Kaveesha	Database Systems	A
Kaveesha	Electronic	A
Sahandi	Numerical Methods	A
Sahandi	Calculas	A
Sahandi	Computer Architecture	A
Sahandi	Software Construction	A
Sahandi	Database Systems	B
Sahandi	Electronic	A

2)

a.Show each course with the number of enrolled students.

```
mysql> SELECT c.course_name, COUNT(e.student_id) AS num_students
    -> FROM courses c
    -> JOIN enrollments e ON c.course_id = e.course_id
    -> GROUP BY c.course_id;
+----------------------+--------------+
| course_name          | num_students |
+----------------------+--------------+
| Numerical Methods    |            6 |
| Calculas             |            6 |
| Computer Architecture |            6 |
| Software Construction |            6 |
| Database Systems     |            6 |
| Electronic           |            6 |
+----------------------+--------------+
6 rows in set (0.01 sec)
```

First group the students by course_id from the enrollments table using GROUP BY. Then use COUNT() to find how many students are enrolled in each course.

3)

a. Find students who enrolled after '2024-02-10'.

```
mysql> SELECT DISTINCT s.*
    -> FROM students s
    -> JOIN enrollments e ON s.student_id = e.student_id
    -> WHERE e.enroll_date > '2024-02-10';
+------------+----------+------------+----------------------+------+
| student_id | name     | dob        | email                | gpa  |
+------------+----------+------------+----------------------+------+
|          6 | Uvindu   | 2002-03-20 | e21006@eng.pdn.ac.lk | NULL |
|         49 | Lahiru   | 2002-09-01 | e21049@eng.pdn.ac.lk | NULL |
|        138 | FIKRY    | 2002-01-15 | e21138@eng.pdn.ac.lk | 3.95 |
|        217 | Ishan    | 2002-06-21 | e21217@eng.pdn.ac.lk | 3.35 |
|        245 | Kaveesha | 2002-06-05 | e21245@eng.pdn.ac.lk | 3.98 |
|        302 | Sahandi  | 2002-05-27 | e21302@eng.pdn.ac.lk | 3.87 |
+------------+----------+------------+----------------------+------+
6 rows in set (0.01 sec)
```

Use WHERE to check if the student's enroll_date is after '2024-02-10' and show only the distinct student details using SELECT DISTINCT.

b. Retrieve names of students who got the highest GPA.

```
mysql> SELECT name
    -> FROM students
    -> WHERE gpa = (SELECT MAX(gpa) FROM students);
+----------+
| name     |
+----------+
| Kaveesha |
+----------+
1 row in set (0.01 sec)
```

Use a subquery to find the highest GPA in the students table, and then retrieve the names of students who have that GPA.

4)

a.Update Lahiru’s GPA to 3.2.

```
mysql> UPDATE students SET gpa = 3.2 WHERE name = 'Lahiru';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM students;
+------------+----------+------------+----------------------+------+
| student_id | name     | dob        | email                | gpa  |
+------------+----------+------------+----------------------+------+
|          6 | Uvindu   | 2002-03-20 | e21006@eng.pdn.ac.lk | NULL |
|         49 | Lahiru   | 2002-09-01 | e21049@eng.pdn.ac.lk | 3.20 |
|        138 | FIKRY    | 2002-01-15 | e21138@eng.pdn.ac.lk | 3.95 |
|        217 | Ishan    | 2002-06-21 | e21217@eng.pdn.ac.lk | 3.35 |
|        245 | Kaveesha | 2002-06-05 | e21245@eng.pdn.ac.lk | 3.98 |
|        302 | Sahandi  | 2002-05-27 | e21302@eng.pdn.ac.lk | 3.87 |
+------------+----------+------------+----------------------+------+
6 rows in set (0.00 sec)
```

Use UPDATE to change the GPA of the student named 'Lahiru' to 3.2 in the students table if the name matches.

b.Delete enrollments where the grade is NULL

```
mysql> DELETE FROM enrollments WHERE grade IS NULL;
Query OK, 4 rows affected (0.01 sec)
```

Use DELETE with WHERE grade IS NULL to remove any enrollment records where the grade is not yet assigned.

5)

a. Create a view named student_course_view showing student name, course name, and grade.

```
mysql> CREATE VIEW student_course_view AS
    -> SELECT s.name AS student_name, c.course_name, e.grade
    -> FROM students s
    -> JOIN enrollments e ON s.student_id = e.student_id
    -> JOIN courses c ON c.course_id = e.course_id;
Query OK, 0 rows affected (1.35 sec)
```

Create a VIEW named student_course_view to simplify access to student names, course names, and grades by using JOIN and SELECT.

```
mysql> SELECT * FROM student_course_view;
+--------------+-----------------------+-------+
| student_name | course_name           | grade |
+--------------+-----------------------+-------+
| Uvindu       | Numerical Methods     | B     |
| Uvindu       | Calculas              | B     |
| Uvindu       | Computer Architecture | B     |
| Uvindu       | Software Construction | B     |
| Uvindu       | Database Systems      | A     |
| Uvindu       | Electronic            | B     |
| Lahiru       | Numerical Methods     | C     |
| Lahiru       | Software Construction | C     |
| FIKRY        | Numerical Methods     | A     |
| FIKRY        | Calculas              | A     |
| FIKRY        | Computer Architecture | A     |
| FIKRY        | Software Construction | A     |
| FIKRY        | Database Systems      | A     |
| FIKRY        | Electronic            | A     |
| Ishan        | Numerical Methods     | B     |
| Ishan        | Calculas              | B     |
| Ishan        | Computer Architecture | B     |
| Ishan        | Software Construction | B     |
| Ishan        | Database Systems      | B     |
| Ishan        | Electronic            | B     |
| Kaveesha     | Numerical Methods     | A     |
| Kaveesha     | Calculas              | A     |
| Kaveesha     | Computer Architecture | A     |
| Kaveesha     | Software Construction | A     |
| Kaveesha     | Database Systems      | A     |
| Kaveesha     | Electronic            | A     |
| Sahandi      | Numerical Methods     | A     |
| Sahandi      | Calculas              | A     |
| Sahandi      | Computer Architecture | A     |
| Sahandi      | Software Construction | A     |
| Sahandi      | Database Systems      | B     |
| Sahandi      | Electronic            | A     |
+--------------+-----------------------+-------+
```

b. Create a stored procedure enroll_student to insert a record into enrollments with student_id, course_id, enroll_date, grade.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE enroll_student (
    ->   IN sid INT, IN cid INT, IN date DATE, IN grd DECIMAL(3,2)
    -> )
    -> BEGIN
    ->   INSERT INTO enrollments (student_id, course_id, enroll_date, grade)
    ->   VALUES (sid, cid, date, grd);
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
```

Create a stored procedure called enroll_student that accepts inputs like student_id, course_id, enroll_date, and grade, and inserts a new record into the enrollments table.